# HARISH SIVARAM

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## THREAT INTELLIGENCE | ADVERSARY TTPs, ANOMALY DETECTION AND AI SECURITY

### PROFESSIONAL SUMMARY

Cybersecurity professional focused on threat intelligence, with an M.Sc. in Cybersecurity and Threat Intelligence (University of Guelph) and 4+ years of software engineering in enterprise production environments. As a Research Associate at eSentire, analyzed attack surfaces in AI agent deployments, including MCP vulnerabilities and the OWASP Top 10 for LLMs, and threat modeled AI architectures with STRIDE. Applied machine learning to threat attribution by classifying malware samples across 43 APT groups from Ghidra-extracted opcode sequences, and built an anomaly detection prototype that flags unauthorized access and code tampering. Programs in Python, C#, and SQL, maps adversary behavior to MITRE ATT&CK, and turns technical findings into clear, actionable guidance for engineering and non-technical stakeholders.

### CORE COMPETENCIES

Threat intelligence and adversary TTP analysis, MITRE ATT&CK mapping, Threat hunting, Machine learning for classification and anomaly detection, Malware analysis and APT attribution, AI and LLM security (MCP, OWASP Top 10 for LLMs), Threat modeling (STRIDE), Data analysis with Python and SQL, Communicating findings to non-technical stakeholders

### WORK EXPERIENCE

**Research Associate** Oct 2025 – Feb 2026

eSentire | Internship | Waterloo, Ontario

- Researched Model Context Protocol (MCP) vulnerabilities and the OWASP Top 10 for LLMs as attack surfaces in AI agent deployments
- Performed STRIDE threat modeling of internal AI architectures, mapping data flows, trust boundaries, and API paths, and reported vulnerabilities with mitigations to engineering teams
- Authored a research paper communicating security findings, risks, and remediation guidance
- Built and documented n8n workflow automation and AI agent orchestration for internal security pipelines, reducing manual effort in repetitive triage tasks, and contributed Terraform and CI/CD updates

**Developer** Apr 2022 – Sep 2024

Doqpros | Full Time | Omaha, Nebraska (Remote)

- Built a Retrieval Augmented Generation (RAG) assistant over product documentation
- Developed C# automation for metadata extraction, validation, and audit logging
- Diagnosed and remediated SSL/TLS certificate and service configuration failures on live client systems, including binding SSL certificates to Laserfiche XML utility endpoints
- Wrote PowerShell scripts to automate routine Windows administration tasks, such as hibernation and power settings
- Supported deployed Laserfiche solutions in client production environments, resolving application, integration, and configuration issues tracked through a ticketing system, and managed Docker, CI/CD, and Azure storage

**Developer** Apr 2020 – Sep 2021

GreenECM Technologies | Full Time | TamilNadu, India

- Installed, configured, and maintained IBM FileNet P8 systems in enterprise environments

### SECURITY TRAINING AND LABS

**SOC Core Skills in the Age of AI** Oct 2026

Antisyphon Training | Instructor-Led Course

**SOC Level 1 Pathway** Mar 2026 – Present

TryHackMe | Self Directed Training

- Hands-on training in cyber defense frameworks, threat intelligence, network traffic analysis, endpoint security monitoring, SIEM, and digital forensics and incident response, with practice in alert triage, investigation, and escalation

**Security Home Lab** Aug 2026 – Present

Self Directed | In Progress

- Running Kali Linux tooling such as nmap against an isolated lab network, driven through an MCP server that connects an AI chat client to the command line, with every action reviewed before it runs

### SELECTED PROJECTS

**APT Malware Classification**, Static Analysis (GitHub)

- Extracted opcode sequences from compiled malware with Ghidra and trained machine learning models to attribute samples to 43 APT groups

**AI Driven Security Chatbot**, Repository Anomaly Detection (Demo)

- Built a prototype that detects anomalous repository activity, flagging unauthorized access and code tampering, and sends real-time alerts with recommended next steps

**EthChat**, Encrypted Messaging (GitHub)

- Developed Solidity smart contracts and message encryption for a censorship-resistant, Ethereum-based messaging app

**MetaWipe**, iOS Privacy Application

- Built a fully offline engine that strips 14+ metadata types, such as GPS and device identifiers, from photos before they are shared

**Pico**, Privacy-First iPhone Budgeting App (nixe.in/pico) **Submitted for App Store Review**

- Designed, tested, and shipped an iOS app with iCloud sync, Siri, and Apple Pay capture, using AI-assisted development (Claude Code) and 140+ automated tests
- Traced silent iCloud sync failures to a Production schema mismatch, fixed it, and added a test and release checklist to prevent a repeat
- Ran a TestFlight beta across 11 builds, triaging tester bug reports from a public Canny feedback board and shipping fixes build by build

**Peer to Peer Energy Trading System**, Blockchain Platform

- Designed the secure transaction framework and automated trading logic for a platform letting households trade energy directly

**Agrimate**, Mobile Finance App (GitHub)

- Built secure backend features for payments and financial tracking in a mobile app improving farmers' financial literacy

## TECHNICAL SKILLS

**Threat Intelligence**: MITRE ATT&CK, adversary TTP analysis, threat hunting, APT attribution, OWASP Top 10, OWASP Top 10 for LLMs, STRIDE

**Data and Machine Learning**: Python, SQL, ML classification and anomaly detection, RAG and LLM applications

**Security Tooling**: Splunk (SIEM), Suricata (IDS), Cisco XDR, Ghidra, Wireshark, nmap, Burp Suite, Metasploit, Kali Linux

**Languages**: Python, C#, SQL, PowerShell, Bash, JavaScript, Swift

**Infrastructure**: Linux, Windows, macOS, Azure, Docker, Terraform, CI/CD, Git, n8n, Kubernetes (familiarity), TCP/IP, DNS, SSL/TLS

## EDUCATION

**Master of Science in Cybersecurity and Threat Intelligence** *(GPA 85.71)*
University of Guelph, Guelph, Ontario Sep 2024 – Aug 2025

- **Relevant Coursework**: Cybersecurity and Defense, Threat Intelligence, Threat Hunting, Penetration Testing, Digital Forensics, Privacy and Compliance (GDPR)

**Bachelor of Engineering in Computer Science and Engineering** *(GPA 9.11/10)*
Sri Krishna College of Engineering and Technology, TamilNadu, India Aug 2019 – May 2023

## CERTIFICATIONS

- Cisco Rapid Incident Response with Cisco XDR
- Cisco Cybersecurity Essentials
- Laserfiche Gold and Platinum Certification 10